

BINOMIAL DISTRIBUTION



SYNOPSIS

Bernoullian Trials (or) Bernoulli Trials

Random trials which result either in the success or failure of an event A, with constant probability of success p and that of failure $1-p=q$ are called as bernoullian trials. For example:

- i) In tossing of an unbiased coin, if we consider getting head upwards as a success then the probability of success $p = \frac{1}{2}$. The probability of failure

$$q = 1 - \frac{1}{2} = \frac{1}{2} \text{ and it is true for every trial.}$$

- ii) In rolling of a symmetrical die, if we consider getting a face with 6 points upward as a success then $p = \frac{1}{6}$

$$\text{and } q = 1 - \frac{1}{6} = \frac{5}{6} \text{ and it is also true for every trial.}$$

Binomial Distribution : The probability of x successes in n independent bernoullian trials is given by $p(X=x) = {}^nC_x p^x q^{n-x}$; $x=0,1,2,3,\dots,n$;

$$p \geq 0, q \geq 0 \text{ and } p+q=1 \text{ and it is called as B.D.}$$

Here n & p are called as the parameters of B.D.

- A discrete random variable x is said to follow B.D. with parameters n, p , if its probability mass function is given by $p(X=x) = {}^nC_x p^x q^{n-x}$; $x=0,1,2,3,\dots,n$; $p \geq 0, q \geq 0$ and $p+q=1$. If X is the number of successes in n trials, then the probability distribution of X is

No. of successes (x)	0	1	...	r		n
$P(X=x)$	${}^nC_0 q^n p^0$	${}^nC_1 q^{n-1} p^1$	...	${}^nC_r q^{n-r} p^r$		${}^nC_n q^0 p^n$

- The above probabilities are various terms of the binomial expansion $(q+p)^n = {}^nC_0 q^n p^0 + {}^nC_1 q^{n-1} p^1 + {}^nC_2 q^{n-2} p^2 + \dots + {}^nC_r q^{n-r} p^r + \dots + {}^nC_n q^0 p^n$
- The originator of B.D. was James Bernoulli (1654-1705) and so it is also some times called as Bernoulli distribution.
- $B(r; n, p)$ means $p(X=r)$

W.E.1:- The value of $B\left(3; 4, \frac{1}{4}\right)$ is

Sol: Here, $r=3, n=4, p=1/4, q=3/4$

$$p(x=3) = {}^4C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^1 = \frac{3}{64}$$

W.E.2:- If the difference between the mean and variance of a binomial distribution for 5 trials is $\frac{5}{9}$, then the distribution is

Sol: Here $n=5, np - npq = 5/9$

$$np(1-q) = 5/9 \Rightarrow 5(p^2) = 5/9; p=1/3, q=2/3 \text{ then, binomial distribution is } (2/3 + 1/3)^5$$

The Characteristics of B.D. :

- The mean of random binomial variate X is np . i.e.,

$$\bar{x} = np \text{ or } np = \bar{x} \therefore n = \frac{\bar{x}}{p}, p = \frac{\bar{x}}{n}$$

- The variance of r.b.v. x is npq . i.e., $\sigma^2 = npq$

$$\text{Now } q = \frac{\sigma^2}{\bar{x}} = \left(\frac{npq}{np}\right)$$

- The S.D. of r.b.v. x is $\sqrt{npq}$ i.e., $\sigma = \sqrt{npq}$

- In binomial distribution $\bar{x} > \sigma^2$ or $\sigma^2 < \bar{x}$ i.e., $np > npq$ or $npq < np$

W.E.3:- A symmetrical die is thrown four times and getting a multiple of 2 is considered to be a success. The mean and variance of success are

Sol : Here, $n=4, p=3/6=1/2, q=1/2$

$$\text{Mean} = np = 4(1/2) = 2$$

$$\text{Variance} = npq = 4(1/2)(1/2) = 1$$

- If $p=q=\frac{1}{2}$, then the distribution is said to be a symmetrical binomial distribution.

MODE : The mode is that value of variable with maximum probability.

- The mode of B.D. depends on the value of $np+p$.

CASE-1: If $np+p=k$, where k is an integer, then there will be two modes namely k & $k-2$. In this case the distribution is said to be a Bi-modal binomial distribution.



CASE-2: If $np+p=k+f$, where k is an integer and f is a proper fraction then there will be only one mode namely k . i.e., the integral part of $np+p$ will be the mode. In this case the distribution is said to be uni-modal binomial distribution.

W.E.4:- In a binomial distribution with $n=10$,

$p = \frac{2}{5}$, the mode of the B.D. is

Sol: Here $np + p = 10 (2/5) + (2/5) = 22/5$
 $= 4 + (2/5)$; Mode = 4

- If we consider n independent bernoullian trials as one experiment and if we repeat such an experiment N times, then the expected frequency or the theoretical frequency of x successes is given by

$$\begin{aligned} f(X=x) &= N.p(X=x) \\ &= N.^nC_x.p^x.q^{n-x}, x=0,1,2,3,...,n \end{aligned}$$

and this is called as Binomial frequency distribution.

W.E.5:- Out of 10,000 families with 4 children each, find the expected number of families all of whose children are daughters

Sol: Here, $p = q = 1/2$, $n = 4$, $N = 10,000$

$$\text{Req.} = N \times P(x=4) = (10,000)^4 C_4 \cdot \left(\frac{1}{2}\right)^4 = 625$$

- In a binomial distribution if ' μ ' is Mean and ' σ ' is S.D. then $0 \leq \sigma < \sqrt{\mu}$ (OR) $[0, \mu)$

W.E.6:- In a binomial distribution $AM=3$, variance=4. The interval of standard deviation is

Sol : Here, Mean (μ) = 3 $\Rightarrow \sigma \in [0, \sqrt{3})$

- In binomial distribution $P(X=r)$ is maximum when $r = \lceil np \rceil$ (where $\lceil \cdot \rceil$ is greatest integer function)

W.E.7:- Suppose X follows binomial distribution

with parameters $n = 100$ and $p = \frac{1}{3}$ then $P(x=r)$ is maximum when r is

Sol: $p(x=r)$ is maximum at $r = [np]$ (where $[]$ is greatest integer function)

$$\Rightarrow r = \left[100 \times \frac{1}{3} \right] = [33.33] = 33$$

W.E.8:- In a binomial distribution $B\left(n, p = \frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to $\frac{9}{10}$, then n is greater than (AIEEE-2009)

Sol: We have $P(x \geq 1) \geq \frac{9}{10} \Rightarrow 1 - P(x = 0) \geq \frac{9}{10}$

$$\Rightarrow 1 - {}^nC_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^n \geq \frac{9}{10} \Rightarrow 1 - \frac{9}{10} \geq \left(\frac{3}{4}\right)^n$$

$$\Rightarrow \left(\frac{3}{4}\right)^n \leq \left(\frac{1}{10}\right)$$

Taking log to the base $3/4$, on both sides, we get

$$n \log_{3/4} \left(\frac{3}{4} \right) \geq \log_{3/4} \left(\frac{1}{10} \right)$$

$$n \geq -\log_{3/4} 10 = \frac{-\log_{10} 10}{\log_{10} \left(\frac{3}{4} \right)} = \frac{-1}{\log_{10} 3 - \log_{10} 4}$$

$$\Rightarrow n \geq \frac{1}{\log_{10} 4 - \log_{10} 3}$$

- 1. The number of parameters of B.D. are**
1) 4 2) 3
3) 2 4) 1
- 2. If $X \sim B(20, 1/2)$. The variance is**
1) 3 2) 4
3) 5 4) 6
- 3. If for a Binomial distribution, $\mu = 10$ and $\sigma^2 = 5$, then $P(X > 6)$ is**

1) $\frac{1}{2^{20}} \sum_{r=7}^{20} {}^{20}C_r$ 2) $\frac{1}{2^{20}} \sum_{r=6}^{20} {}^{20}C_r$
3) $\frac{1}{2^{19}} \sum_{r=7}^{20} {}^{20}C_r$ 4) $\frac{1}{2^{19}} \sum_{r=6}^{20} {}^{20}C_r$
- 4. Let X be a $B(2, p)$ and Y be an independent $B(4, p)$. If $P(X \geq 1) = 5/9$, then $P(Y \geq 1)$ is**
1) $\frac{61}{65}$ 2) $\frac{65}{81}$ 3) $\frac{1}{2}$ 4) $\frac{1}{3}$

- When a coin is tossed n times if the probability for getting 6 heads is equal to the probability for getting 8 heads then, the value of n is
1) 8 2) 10 3) 12 4) 14
- The least number of times a fair coin must be tossed so that the probability of getting atleast one head is atleast 0.8 is
1) 1 2) 2 3) 3 4) 4
- In a box containing 15 identical bulbs, 5 are defective. If 5 bulbs are drawn at random from the box with replacement, The probability that atleast one of them is defective is
1) $\frac{32}{343}$ 2) $\frac{211}{243}$ 3) $\frac{80}{243}$ 4) $\frac{32}{243}$
- When a coin is tossed, $p(X = r \text{ heads}) =$
1) $\frac{{}^n C_r}{2^n}$ 2) $\frac{{}^n C_r}{2^{n-1}}$ 3) $\frac{{}^n C_n}{2^{n-1}}$ 4) $\frac{{}^n C_n}{2^n}$

C.U.Q. - KEY

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|------|------|------|------|
| 1) 3 | 2) 3 | 3) 1 | 4) 2 |
| 5) 4 | 6) 3 | 7) 2 | 8) 1 |

C.U.Q. - HINTS

- n & p
- $X \sim B(20, 1/2)$
 $\Rightarrow n = 20, p = 1/2, q = 1/2$
mean $= np = 10$
variance $= npq = 20(1/2)(1/2) = 5$
- $P(X > 6) = \sum_{r=7}^{20} {}^{20}C_r \left(\frac{1}{2}\right)^{20-r} \left(\frac{1}{2}\right)^r = \frac{1}{2^{20}} \sum_{r=7}^{20} {}^{20}C_r$
- $P(X = x) = {}^2 C_x p^x q^{2-x}$; $P(Y = y) = {}^4 C_y p^y q^{4-y}$
 $P(X \geq 1) = 1 - P(X = 0) = 1 - q^2 = 5/9$
 $\Rightarrow q = 2/3, p = 1/3$
 $P(Y \geq 1) = 1 - P(Y = 0) = 1 - q^4 = 65/81$
- ${}^n C_6 = {}^n C_8 \Rightarrow n = 14$
- $1 - \frac{1}{2^n} \geq 0.8$
- $P(\text{atleast one of them is defective})$
 $= P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{32}{243} = \frac{211}{243}$
- $p = q = \frac{1}{2}$



LEVEL-I (C.W)

- A binomial distribution has a mean of 5 and variance 4. The number of trials is
1) 20 2) 16 3) 25 4) 10
- If X is binomial variate with $E(X) = 5$ and variance 4. The parameters of the distribution are
1) $\frac{1}{4}, 20$ 2) $\frac{1}{5}, 20$ 3) $25, \frac{1}{5}$ 4) $\frac{1}{25}, \frac{1}{5}$
- If the mean of the binomial distribution is 25. Then standard deviation lies in the interval given below (Eamcet 1992)
1) $[0, 5)$ 2) $(0, 5]$ 3) $[0, 25)$ 4) $(0, 25]$
- The standard deviation σ of $(q + p)^{16}$ is 2. The mean of the distribution is
1) 2 2) 8 3) 16 4) 20
- The probability of getting a total of 9 atleast twice in 6 tosses of a pair of dice is
1) ${}^6 C_2 \left(\frac{1}{9}\right)^2 \left(\frac{8}{9}\right)^4$ 2) $1 - \left\{ \sum_{x=2}^6 {}^6 C_x \left(\frac{1}{9}\right)^x \left(\frac{8}{9}\right)^{6-x} \right\}$
3) $\sum_{x=2}^6 {}^6 C_x \left(\frac{1}{9}\right)^x \left(\frac{8}{9}\right)^{6-x}$ 4) $\sum_{x=2}^6 {}^6 C_x \left(\frac{8}{9}\right)^x \left(\frac{1}{9}\right)^{6-x}$
- In a binomial distribution, if AM = 4 and variance = 3. Then n is
1) 10 2) 12 3) 4 4) 3
- In a binomial distribution mean is 4.8 and variance is 2.88, then the parameter n is
1) 8 2) 12 3) 16 4) 20
- A fair coin is tossed 6 times. The variance of number of heads is
1) 3 2) $3/2$ 3) $3/4$ 4) 2
- In a binomial distribution mean $= \frac{11}{4}$ and variance $= \frac{15}{16}$, then the probability of success is
1) $\frac{1}{2}$ 2) $\frac{29}{44}$ 3) $\frac{1}{4}$ 4) $\frac{3}{4}$
- A machine manufacturing screws is known to produce 5% defectives. In a random sample of 15 screws the probability that there are exactly 3 defectives is
1) $\frac{16}{20}$ 2) $\frac{17}{20}$ 3) ${}^{15}C_3 \left(\frac{1}{20}\right)^3 \left(\frac{19}{20}\right)^{12}$ 4) $\frac{18}{20}$



11. The probability of a man hitting the target is $\frac{1}{4}$. If he fires 7 times the probability of his hitting the target atleast once is
 1) $\left(\frac{3}{4}\right)^7$ 2) $1 - \left(\frac{3}{4}\right)^7$ 3) $\left(\frac{1}{4}\right)^7$ 4) $1 - \left(\frac{1}{4}\right)^7$
12. A box contains 3 red marbles and 2 white marbles. A marble is drawn and replaced three times from the box. The probability that exactly one red marble is drawn is
 1) $3/5$ 2) $9/125$ 3) $36/125$ 4) $6/125$
13. Of the bolts produced by a factory 2% are defective. In a shipment of 3600 bolts from the factory, the expected number of defective bolts is
 1) 144 2) 72 3) 36 4) 18
14. If X is a binomial variate with $n=6$ and $9P(X=4)=P(X=2)$, the parameter p is
 1) $3/4$ 2) $1/3$ 3) $1/4$ 4) $1/2$
15. Twenty identical coins each with probability p of showing heads are tossed. The probability of heads showing on 10 coins is same as that of heads showing on 11 coins then p =
 1) $\frac{1}{2}$ 2) $\frac{10}{21}$ 3) $\frac{11}{21}$ 4) $\frac{13}{21}$
16. The probability that a student is not a swimmer is $\frac{1}{5}$. Out of 5 students the probability that exactly four are swimmers is
 1) $\left(\frac{4}{5}\right)^3$ 2) $\left(\frac{1}{5}\right)^4$
 3) ${}^5C_4 \left(\frac{4}{5}\right)^4 \left(\frac{1}{5}\right)^1$ 4) ${}^5C_4 \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right)^1$
17. The probability that India wins a cricket test match against England is $\frac{1}{3}$. If India and England play 3 matches, the probability that India will win atleast one match is
 1) $8/27$ 2) $19/27$ 3) $1/27$ 4) $9/27$
18. In a binomial distribution, the probability of getting a success is $\frac{1}{4}$ and the standard deviation is 3. Then its mean is
 1) 12 2) 10 3) 8 4) 6
19. The probability of obtaining 2 heads when an unbiased coin is tossed 5 times is
 1) $5/8$ 2) $4/9$ 3) $5/16$ 4) $4/16$
20. Six unbiased coins are tossed, the probability of obtaining atleast two heads is
 1) $50/64$ 2) $55/64$ 3) $57/64$ 4) $60/64$
21. The probability of answering 6 out of 10 questions correctly in a true or false examination is
 1) ${}^{10}C_4 \left(\frac{1}{2}\right)^4$ 2) ${}^{10}C_6 \left(\frac{1}{2}\right)^6$ 3) ${}^{10}C_6 \left(\frac{1}{2}\right)^{10}$ 4) ${}^{10}C_6 \left(\frac{1}{2}\right)^8$
22. On the average if it rains on 5 days in every 30 days, the probability that there will be rain on exactly three days of a given week is
 1) ${}^7C_3 \left(\frac{1}{6}\right)^3$ 2) ${}^7C_3 \left(\frac{5}{6}\right)^3$
 3) ${}^7C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^4$ 4) ${}^7C_3 \left(\frac{1}{6}\right)^4$
23. A and B play a game in which A's chance of winning is $\frac{1}{5}$. In a series of 6 games, the probability that A will win all the 6 games is
 1) ${}^6C_2 \left(\frac{1}{5}\right)^6$ 2) ${}^6C_6 \left(\frac{1}{5}\right)^6 \left(\frac{4}{5}\right)^0$
 3) $\left(\frac{4}{5}\right)^6$ 4) ${}^6C_5 \left(\frac{1}{5}\right)^5$
24. The probability that in a family of 4 children there will be atleast one boy is
 1) $\frac{13}{16}$ 2) $\frac{15}{16}$ 3) $\frac{14}{16}$ 4) $\frac{12}{16}$
25. A fair coin is tossed four times. The probability that tails exceed heads in number is
 1) $1/4$ 2) $5/16$ 3) $7/16$ 4) $9/16$
26. A symmetrical die is thrown 6 times. If getting an odd number is a success, the probability of at the most 5 successes is
 1) $\frac{5}{64}$ 2) $\frac{15}{64}$ 3) $\frac{63}{64}$ 4) $\frac{36}{64}$
27. Five cards are drawn successively with replacement from a well shuffled pack of 52 cards. The probability that all the five cards are spades
 1) ${}^5C_5 \left(\frac{1}{4}\right)^5$ 2) $\frac{5}{52}$ 3) $\left(\frac{3}{4}\right)^5$ 4) $\left(\frac{3}{4}\right)^2$

28. In six throws of a die, getting 4 or 5 is considered as a success. The mean number of successes is

- 1) 4 2) 3 3) 2 4) 1

29. If A and B are two equally strong table tennis players, the probability that A beats B in exactly three games out of 4 games is

- 1) $\frac{1}{6}$ 2) $\frac{1}{5}$ 3) $\frac{1}{4}$ 4) $\frac{1}{2}$

30. A student is given 6 questions in a true or false examination. If he gets 4 or more correct answers, he passes the examination. The probability that he passes the examination is

- 1) $\frac{5}{32}$ 2) $\frac{7}{32}$ 3) $\frac{11}{32}$ 4) $\frac{3}{32}$

31. A machine manufacturing screws is known to produce 5% defectives. In a random sample of 15 screws the probability that there are not more than 3 defectives is

- 1) ${}^{15}C_3 \left(\frac{1}{20}\right)^{12}$ 2) $\sum_{x=4}^{15} {}^{15}C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{15-x}$
 3) $\sum_{x=0}^3 {}^{15}C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{15-x}$ 4) $\sum_{x=0}^3 {}^{15}C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{15}$

32. The incidence of occupational disease in an industry is such that the workers have a 20% chance of suffering from it. The probability that out of 6 workers chosen at random, not even one will suffer from that disease is

- 1) $\left(\frac{1}{5}\right)^6$ 2) $\left(\frac{4}{5}\right)^6$ 3) $\left(\frac{1}{5}\right)^0$ 4) $\left(\frac{1}{5}\right)^3$

33. The chance that a person with two dice, the faces of each being numbered 1 to 6, will throw aces exactly 4 times in 6 trials is

- 1) $\left(\frac{1}{36}\right)^4$ 2) $\left(\frac{35}{36}\right)^4$
 3) ${}^6C_4 \left(\frac{1}{36}\right)^4 \left(\frac{35}{36}\right)^2$ 4) ${}^6C_4 \left(\frac{35}{36}\right)^4 \left(\frac{1}{36}\right)^2$

34. If 10% of the attacking air crafts are expected to be shot down before reaching the target, the probability that out of 5 aircrafts atleast four will be shot before they reach the target is

- 1) $4\left(\frac{1}{10}\right)^4$ 2) $5\left(\frac{1}{10}\right)^4$ 3) $(4.6)\left(\frac{1}{10}\right)^4$ 4) $6\left(\frac{1}{10}\right)^4$

35. The probability of A winning a game is $\frac{2}{3}$. In a series of 4 such games the probability that A will win more than half of the games is

- 1) $\frac{32}{81}$ 2) $\frac{33}{81}$ 3) $\frac{16}{27}$ 4) $\frac{8}{27}$

LEVEL-I (C.W.) - KEY

- 1) 3 2) 3 3) 1 4) 2 5) 3 6) 2 7) 2
 8) 2 9) 2 10) 3 11) 2 12) 3 13) 2 14) 3
 15) 3 16) 3 17) 2 18) 1 19) 3 20) 3 21) 3
 22) 3 23) 2 24) 2 25) 2 26) 3 27) 1 28) 3
 29) 3 30) 3 31) 3 32) 2 33) 3 34) 3 35) 3

LEVEL-I (C.W.) - HINTS

- $np = 5, npq = 4$
- $np = 5, npq = 4, \text{Req} = n, p$
- $np = 25, \text{S.D} = [0, \sqrt{\mu}]$
- $npq = 4, n = 16$
- $P(x \geq 2)$
- $np = 4, npq = 3$
- $np = 4.8, npq = 2.88$
- $n = 6, \text{Req.} = npq = 6 \times \frac{1}{2} \times \frac{1}{2}$
- $\text{Req.} = p$
- $n = 15, p = \frac{5}{100}, P(X = 3)$
- $p = \frac{1}{4}, n = 7, P(X \geq 1)$
- $n = 3, p(x = 1), p = \frac{3}{5}$
- $p = \frac{2}{100}, n = 3600, \mu = np$
- $p(x = r) = {}^nC_r q^{n-r} \cdot p^r$
- $n = 20, P(x = 10) = P(x = 11)$
- $q = \frac{1}{5}, n = 5, p(x = 4)$
- $p = \frac{1}{3}, n = 3, p(x \geq 1)$
- $p = \frac{1}{4}, npq = 9$
- $n = 5, p = q = \frac{1}{2}, p(x = 2)$
- $n = 6, P(X \geq 2)$
- $n = 10, p = q = \frac{1}{2}, r = 6$
- $P(X = 3), n = 7, p = \frac{5}{30}$

23. $p = \frac{1}{5}, n = 6, P(X = 6)$
24. $n = 4, p(x \geq 1), p = q = \frac{1}{2}$
25. $\text{Req.} = \frac{{}^4C_3 + {}^4C_4}{2^4}$
26. $P(X \leq 5), n = 6, p = \frac{3}{6}$
27. $n = 5, p = \frac{13}{52}$
28. $n = 6, p = \frac{2}{6}, \text{Req} = np$
29. $p(x = 3), n = 4, p = q = \frac{1}{2}$
30. $n = 6, p(x \geq 4), p = q = \frac{1}{2}$
31. $P(x \leq 3), n = 15, P = \frac{5}{100}$
32. $P(x < 1), n = 6, p = 20\%$
33. $n = 6, p(x = 4), p = \frac{1}{36}$
34. $n = 5, p(x \geq 4), p = \frac{10}{100}$
35. $p = \frac{2}{3}, q = \frac{1}{3}, n = 4, p(x > 2)$



LEVEL-I (H.W)

1. The mean of binomial distribution is 6 and its S.D. is $\sqrt{2}$, then the number of trials n is
1) 7 2) 8 3) 9 4) 10
2. If a random variable X follows B.D. with mean 2.4 and variance 1.44, the number of independent trials n is
1) 10 2) 8 3) 6 4) 2
3. If for a binomial distribution with $n = 5$, $4p(X=1)=P(X=2)$, the probability of success is
1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{8}$
4. A symmetrical die is rolled 720 times. Getting a face with four points is considered to be a success. The mean and variance of the number of successes is
1) 20, 120 2) 120, 100 3) 100, 100 4) 50, 50

5. The probability of getting atleast two heads when an unbiased coin is tossed three times is
1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{8}$
6. For a binomial distribution $\bar{x} = 4, \sigma^2 = 3$ then the distribution of x is
1) $\left(\frac{1}{4} + \frac{3}{4}\right)^{16}$ 2) $\left(\frac{3}{4} + \frac{1}{4}\right)^{16}$ 3) $\left(\frac{1}{2} + \frac{1}{2}\right)^{16}$ 4) $\left(\frac{1}{2} + \frac{1}{2}\right)^8$
7. If for a binomial distribution mean $= \frac{10}{3}$ and sum of mean and variance is $\frac{40}{9}$. The parameters are
1) $\frac{2}{3}, 10$ 2) $\frac{2}{3}, 20$ 3) $5, \frac{2}{3}$ 4) $4, \frac{2}{3}$
8. If for a binomial distribution $\bar{x} = \frac{6}{5}$ and the difference between mean and variance is $\frac{6}{25}$. The number of trials is
1) 8 2) 7 3) 6 4) 5
9. If for a B.D. with $n=12$, the ratio of variance to mean is $\frac{1}{3}$, the probability of 10 successes is
1) ${}^{15}C_{10} \left(\frac{2}{3}\right)^{10} \left(\frac{1}{3}\right)^2$ 2) ${}^{12}C_{10} \left(\frac{2}{3}\right)^{10} \left(\frac{1}{3}\right)^2$
3) $\left(\frac{2}{3}\right)^{10}$ 4) $\left(\frac{1}{3}\right)^{10}$
10. A symmetrical die is thrown three times. If getting a six is considered to be a success, the probability of atleast two successes is
1) $\frac{4}{27}$ 2) $\frac{3}{27}$ 3) $\frac{2}{27}$ 4) $\frac{1}{27}$
11. If for a BD the mean is 6 and standard deviation is $\frac{1}{\sqrt{2}}$, the probability of success is
1) 11/12 2) 10/12 3) 9/12 4) 8/12
12. If the sum of mean and variance of B.D. for 5 trials is 1.8, the binomial distribution is
1) $(0.8 + 0.2)^5$ 2) $(0.2 + 0.8)^5$
3) $(0.8 + 0.2)^{10}$ 4) $(0.2 + 0.8)^{10}$



13. If for a binomial distribution $n = 4$ and $6P(X=4)=P(X=2)$, the probability of success is
1) $3/4$ 2) $1/2$ 3) $1/3$ 4) $1/4$
14. A box contains 6 red and 4 white marbles. A marble is drawn and replaced three times from the box. The probability that one white marble is drawn
1) $\frac{53}{125}$ 2) $\frac{54}{125}$ 3) $\frac{56}{125}$ 4) $\frac{52}{125}$
15. A random variable X is binomially distributed with mean is 12 and variance is 8. Find the parameters of the distribution are
1) $18, \frac{1}{3}$ 2) $36, \frac{1}{3}$ 3) $36, \frac{2}{3}$ 4) $18, \frac{2}{3}$
16. If the mean and variance of a binomial variate X are 8 and 4 respectively then $P(X < 3) =$ (EAM-2014)
1) $\frac{137}{2^{16}}$ 2) $\frac{697}{2^{16}}$ 3) $\frac{265}{2^{16}}$ 4) $\frac{265}{2^{15}}$
17. The mean and standard deviation of a binomial variate X are 4 and $\sqrt{3}$ respectively. Then $P(X \geq 1) =$ (EAM-2007)
1) $1 - \left(\frac{1}{4}\right)^{16}$ 2) $1 - \left(\frac{3}{4}\right)^{16}$ 3) $1 - \left(\frac{2}{3}\right)^{16}$ 4) $1 - \left(\frac{1}{3}\right)^{16}$
18. If the mean and variance of a binomial variable X are 2 and 1 respectively, then $P(X \geq 1) =$ (EAM-2010)
1) $\frac{2}{3}$ 2) $\frac{15}{16}$ 3) $\frac{7}{8}$ 4) $\frac{4}{5}$
19. X follows a binomial distribution with parameters $n = 6$ and p . If $4P(X = 4) = P(X = 2)$. Then $p =$ (EAM-2009)
1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{1}{6}$ 4) $\frac{1}{3}$
20. Suppose X follows a binomial distribution with parameters n and p , where $0 < p < 1$. If $\frac{P(X=r)}{P(X=n-r)}$ is independent of n for every r , then $p =$ (EAM-2012)
1) $1/2$ 2) $1/3$ 3) $1/4$ 4) $1/8$

LEVEL-I (H.W.) - KEY

- | | | | |
|-------|-------|-------|-------|
| 1) 3 | 2) 3 | 3) 2 | 4) 2 |
| 5) 3 | 6) 2 | 7) 3 | 8) 3 |
| 9) 2 | 10) 3 | 11) 1 | 12) 1 |
| 13) 2 | 14) 2 | 15) 2 | 16) 1 |
| 17) 2 | 18) 2 | 19) 4 | 20) 1 |

LEVEL-I (H.W.) - HINTS

- $np = 6, npq = 2$
- $np = 2.4, npq = 1.44$
- $n = 720, p = \frac{1}{6}, q = \frac{5}{6}$
- $n = 3, P(X \geq 2)$
- $np = 4, npq = 3$
- Req. = n, p
- Req. = n
- $P(x = 10)$
- $n = 3, p = \frac{1}{6}, P(x \geq 2)$
- $np = 6, npq = \frac{1}{2}$
- $np + npq = 1.8, n = 5$
- Req. = p
- $n = 6, p(x = 4), p = \frac{1}{36}$
- $n = 3, p = \frac{4}{10}, P(X = 1)$
- Req. = n, p
- $p(x < 3) = \frac{1}{2^{16}} ({}^{16}C_0 + {}^{16}C_1 + {}^{16}C_2)$
- $q = \frac{3}{4}, p = \frac{1}{4}, n = 16$
- $p = q = \frac{1}{2}, n = 4$
- $p = \frac{1}{3}$
- $\frac{{}^nC_r p^r q^{n-r}}{{}^nC_r p^{n-r} q^r}$ is independent of n
 $p = q = 1/2$



LEVEL-II (C.W)

- For a binomial distribution if $p = \frac{1}{4}, n = 20$ the probability of mode is
 1) ${}^{20}C_5 \left(\frac{3}{4}\right)^5$ 2) ${}^{20}C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^{15}$
 3) ${}^{10}C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{10}$ 4) ${}^{10}C_{10} \left(\frac{3}{4}\right)^{10}$
- In a B.D. $n = 400, p = \frac{1}{5}$. Its standard deviation is
 1) $10 \times \sqrt{2}$ 2) $\frac{1}{800}$ 3) 4 4) 8
- If X be B.V. with $E(X) = 5$ and $E(X^2) - \{E(X)\}^2 = 4$, then the parameters of distribution are
 1) $\frac{1}{4}, 20$ 2) $\frac{1}{5}, 20$ 3) $\frac{1}{5}, 25$ 4) $\frac{4}{5}, 25$
- If x is $B\left(x, n, \frac{1}{3}\right), P(x \geq 1) > 0.8$, the least value of n is
 1) 3 2) 4 3) 5 4) 6
- A box contains 'a' white and 'b' black balls. 'c' balls are drawn at random with replacement. The expected number of white balls drawn is
 1) $\frac{a}{a+b}$ 2) $\frac{ac}{a+b}$ 3) $\frac{bc}{a+b}$ 4) $\frac{b}{a+b}$
- 5 cards are drawn one after another successively with replacement from a well shuffled pack of 52 cards. The probability that all the 5 cards are spades
 1) $\left(\frac{3}{4}\right)^5$ 2) $1 - \left(\frac{3}{4}\right)^5$ 3) $\left(\frac{1}{4}\right)^5$ 4) $1 - \left(\frac{1}{4}\right)^5$
- A card is drawn and replaced four times from an ordinary pack of 52 playing cards. The probability that at least once heart is drawn
 1) $\left(\frac{3}{4}\right)^4$ 2) $1 - \left(\frac{1}{2}\right)^4$
 3) $1 - \left(\frac{3}{4}\right)^4$ 4) $\left(\frac{1}{2}\right)^4$
- Out of 800 families with 4 children each the expected number of families having 2 boys and 2 girls is
 1) 100 2) 200 3) 300 4) 400
- One hundred identical coins each with probability P of showing up heads are tossed. If $0 < P < 1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins then the value of P is
 1) $\frac{50}{100}$ 2) $\frac{51}{101}$ 3) $\frac{52}{101}$ 4) $\frac{53}{101}$
- $2K+1$ coins (K is an integer) each with probability P ($0 < P < 1$) of getting head are tossed together. If the probability of getting K heads is equal to the probability of getting $K+1$ heads, the value of P is
 1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{8}$
- A card is drawn and replaced in an ordinary pack of playing cards. The number of times a card must be drawn so that the probability of getting atleast a club card is greater than $\frac{3}{4}$
 1) 7 2) 6 3) 5 4) 4
- Out of 2000 families with 4 children, the number of familes you expect to have atleast one boy is
 1) 1875 2) 750 3) 1250 4) 625
- 6 symmetrical dice are thrown 1458 times. The number of times you expect three dice to show a four or five is
 1) 160 2) 320 3) 480 4) 600
- If a sex ratio of births is 49 girls to 51 boys, the probability that there will be 8 girls amongst 10 babies born on the same day in a maternity hospital is
 1) ${}^{10}C_8 (0.51)^8 (0.49)^2$ 2) ${}^{10}C_8 (0.49)^8 (0.51)^2$
 3) ${}^{10}C_8 (0.49 \times 0.51)^8$ 4) ${}^{10}C_8 (0.49 \times 0.51)^2$
- In a market region half of the households is known to use a particular brand of soap. In a household survey, a sample of 10 house holds are allotted to each investigator and 2048 investigators are appointed for the survey. The number of investigators likely to report that there are three households is
 1) 240 2) 352 3) 1696 4) 120



16. In a market region half of the households is known to use a particular brand of soap. In a household survey, a sample of 10 households are allotted to each investigator and 2048 investigators are appointed for the survey. The number of investigators likely to report that there are atleast 4 users is
1) 240 2) 352 3) 1696 4) 120
17. The least number of times a fair coin is to be tossed in order that the probability of getting atleast one head is at least 0.99 is
1) 5 2) 6 3) 7 4) 8
18. An arcade game is such that the probability of any person winning is always 0.3. The minimum number of people play the game to ensure that the probability that atleast one person wins is greater than or equal to 0.96 is
1) 8 2) 9 3) 10 4) 12
19. Suppose A and B are two equally strong table tennis players. Which of the following two events is more probable
a) A beats B in exactly 3 games out of 4
b) A beats B in exactly 5 games out of 8
1) a 2) b
3) a & b 4) neither a nor b
20. A bag contains 13 balls numbered from 1 to 13. Suppose drawing of an even number is a success. Two balls are drawn with replacement from the bag. The probability of getting two successes is
1) $\frac{84}{169}$ 2) $\frac{49}{169}$ 3) $\frac{36}{169}$ 4) $\frac{120}{169}$
21. A coin tossed n times. If the probability that 4, 5, 6 heads occur are in A.P., then $n =$
1) 14 2) 8 3) 15 4) 11
22. Suppose X follows binomial distribution with parameters $n = 100$ and $p = \frac{1}{8}$ then $P(X=r)$ is maximum when $r =$
1) 49 2) 50 3) 12 4) 34
23. The probability of a bomb hitting a bridge is $\frac{1}{2}$ and two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9 is
1) 5 2) 6 3) 8 4) 7

24. A coin whose faces are marked 3 & 5 is tossed 4 times. The probability that the sum of the numbers thrown is greater than 15
1) $\frac{11}{16}$ 2) $\frac{5}{16}$ 3) $\frac{5}{8}$ 4) $\frac{1}{16}$
25. The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is
1) $128/256$ 2) $219/256$
3) $37/256$ 4) $28/256$
26. If the mean and the variance of a binomial variate X are 2 and 1 respectively, then the probability that X takes a value greater than one is equal to
1) $\frac{1}{16}$ 2) $\frac{5}{16}$ 3) $\frac{11}{16}$ 4) $\frac{15}{16}$

LEVEL-II (C.W.) - KEY

- 1) 2 2) 4 3) 3 4) 2 5) 2 6) 3 7) 3
8) 3 9) 2 10) 3 11) 3 12) 1 13) 2 14) 2
15) 1 16) 3 17) 3 18) 3 19) 1 20) 3 21) 1
22) 3 23) 4 24) 1 25) 4 26) 3

LEVEL-II (C.W.) - HINTS

1. $p = \frac{1}{4}, n = 20, np + p = 20 \times \frac{1}{4} + \frac{1}{4} = 5 + \frac{1}{4}$
i.e., mode = 5
probability of mode = $P(X=5) = {}^{20}C_5 \left(\frac{3}{4}\right)^{15} \left(\frac{1}{4}\right)^5$
2. S.D. = $\sqrt{npq}$
3. $E(x) = np = 5, E(x^2) - [E(x)]^2 = npq = 4$
4. $P(X \geq 1) > 0.8 \Rightarrow 1 - P(X=0) > 0.8$
 $\Rightarrow \left(\frac{2}{3}\right)^n < 0.2 \Rightarrow$ Least value of n is 4
5. $\mu = np$
6. $P(X=5), n=5, p=\frac{13}{52}$
7. $P(X \geq 1), n=4, p=\frac{13}{52}$
8. $\text{Req.} = N \times P(X=2), N=800, p=q=\frac{1}{2}, n=4$
9. $n=100, P(X=50) = P(X=51)$
10. $n=2k+1, P(X=k) = P(X=k+1)$

11. $P(x \geq 1) > \frac{3}{4}$

$$1 - P(X=0) > \frac{3}{4}, p = \frac{13}{52}$$

12. $\text{Req.} = N \times P(\geq 1)$

$$N = 2000, n = 4, p = q = \frac{1}{2}$$

13. $\text{Req.} = 1458 \times P(x=3), n=6, p = \frac{2}{6}$

14. $p = \frac{49}{100}, n=10, P(X=8)$

15. $p = \frac{1}{2}, n=10, N=2048$

$$N \cdot P(X=3) = 2048 \times \frac{{}^{10}C_3}{2^{10}} = 240$$

16. $2048 - 352 = 1696$

17. $P(X \geq 1) \geq 0.99$

18. $1 - (0.7)^n \geq 0.96$

$$0.04 > (0.7)^n$$

19. $P(E_1) = \frac{{}^4C_3}{2^4} = \frac{4}{16} = \frac{8}{32}$

20. $P(X=2), n=2$
even = $\{2, 4, 6, 8, 10, 12\}$

$$p = \frac{6}{13}$$

21. ${}^nC_4, {}^nC_5, {}^nC_6$ are in A.P. $(n-2r)^2 = n+2$, where
 $r=5$

22. $p(x=r)$ is maximum at $r = [np]$

23. $p = \frac{1}{2}, q = \frac{1}{2}, P(x \geq 2) > 0.9 \Rightarrow (n+1)\frac{1}{2^n} < \frac{1}{10}$

24. $\begin{matrix} 16 & 18 & 20 \\ (5533) & (5553) & (5555) \end{matrix}$

$$n=4, p=q = \frac{1}{2}$$

$$\text{Req.} = p(X=2) + p(X=3) + p(X=4)$$

25. $np=4, npq=2, P(x=2)$

26. $p=q = \frac{1}{2}, n=4$



LEVEL-II (H.W)

- The mean and the variance of a random variable X having a binomial distribution are 4 and 2 respectively, then $P(X=1)$ is
1) $\frac{1}{32}$ 2) $\frac{1}{16}$ 3) $\frac{1}{8}$ 4) $\frac{1}{4}$
- A card is drawn and replaced three times from an ordinary pack of 52 playing cards. The probability that three spades are drawn is
1) $\frac{9}{64}$ 2) $\frac{1}{64}$ 3) $\frac{37}{64}$ 4) $\frac{3}{64}$
- If we take 1280 sets each of 10 tosses of a fair coin, the number of sets we expect to get 7 heads and 3 tails is
1) 450 2) 300 3) 150 4) 75
- 201 coins each with probability $P(0 < P < 1)$ of showing head are tossed together. If the probability of getting 100 heads is equal to the probability of getting 101 heads, then the value of p is
1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{6}$
- The probability of a man hitting the target is $\frac{1}{4}$. The number of times atleast must he fire so that the probability of his hitting the target atleast once is greater than $\frac{2}{3}$ is
1) 6 2) 5 3) 4 4) 3
- A bag contains 13 balls numbered from 1 to 13. Suppose drawing of an even number is a success. Two balls are drawn with replacement from the bag. The probability of getting atleast one success is
1) $\frac{49}{169}$ 2) $\frac{84}{169}$ 3) $\frac{120}{169}$ 4) $\frac{36}{169}$
- One in 9 ships is likely to be wrecked, when they are set on sail. When 6 ships set on sail, the probability for exactly, 3 will not arrive safely is

$$1) \frac{25 \times 8^3}{9^6} \quad 2) 1 - \frac{1}{9^6} \quad 3) {}^6C_3 \left(\frac{8^3}{9^6} \right) \quad 4) {}^6C_3 \left(\frac{8^6}{9^3} \right)$$

LEVEL-II (H.W.) - KEY

- 1) 1 2) 2 3) 3 4) 3
5) 3 6) 3 7) 3

LEVEL-II (H.W.) - HINTS

1. $np = 4, npq = 2, P(x = 1)$
2. $P(X = 3), n = 3, p = \frac{13}{52}$
3. $\text{Req.} = 1280 \times p(X = 7), n = 10, p = q = \frac{1}{2}$
4. $n = 201, P(X = 100) = P(X = 101)$
5. $P(X \geq 1) > \frac{2}{3}, p = \frac{1}{4}$
6. $P(X \geq 1), n = 2, p = \frac{6}{13}$
7. $p = \frac{1}{9}, q = \frac{8}{9}, n = 6$



LEVEL - III

1. The largest possible variance of a binomial variate is
 - 1) n 2) $\frac{n}{2}$ 3) $\frac{n}{4}$ 4) $\frac{n}{6}$
2. Five coins whose faces are marked 2 and 3 are thrown. The chance of obtaining a total of 12 is
 - 1) $\frac{11}{16}$ 2) $\frac{15}{16}$ 3) $\frac{5}{16}$ 4) $\frac{1}{16}$
3. An experiment succeeds twice as often as it fails. The chance that in the next six trials, there shall be atleast four successes is
 - 1) ${}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2$ 2) $\sum_{x=0}^4 {}^6C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{6-x}$
 - 3) $\sum_{x=4}^6 {}^6C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{6-x}$ 4) $\sum_{x=6}^8 {}^6C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{6-x}$
4. The probability that a bulb produced by a factory will fuse after 100 days of use is 0.05. The probability that out of 5 such bulbs atleast one will fuse after 100 days of use is
 - 1) $(0.95)^5$ 2) $(0.05)^5$ 3) $1 - (0.95)^5$ 4) $1 - (0.05)^5$
5. The probability that a bulb produced by a factory will fuse after 100 days of use is 0.05. The probability that out of 5 such bulbs none of them fuse after 100 days is
 - 1) $(0.05)^5$ 2) $(0.95)^5$
 - 3) $1 - (0.95)^5$ 4) $1 - (0.05)^5$

6. A coin whose faces are marked 3 & 5 is tossed 4 times. The probability that the sum of the numbers thrown is 12 is
 - 1) $\frac{1}{16}$ 2) $\frac{5}{16}$ 3) $\frac{5}{8}$ 4) $\frac{1}{8}$
7. The sum and product of mean and variance of a binomial distribution are 24 and 128 respectively. The binomial distribution is
 - 1) $\left(\frac{1}{2} + \frac{1}{2}\right)^{32}$ 2) $\left(\frac{3}{10} + \frac{7}{10}\right)^{32}$
 - 3) $\left(\frac{1}{50} + \frac{49}{50}\right)^{32}$ 4) $\left(\frac{1}{3} + \frac{2}{3}\right)^{32}$
8. X and Y are independent binomial variates $B\left(5, \frac{1}{2}\right)$ and $B\left(7, \frac{1}{2}\right)$ then $P(X + Y = 3)$ is
 - 1) $\frac{45}{1024}$ 2) $\frac{55}{1024}$ 3) $\frac{65}{1024}$ 4) $\frac{60}{1024}$
9. A die is tossed twice. Getting 'an odd number' is termed a success. The probability distribution of number of successes (X) is formed. Then its mean, variance are
 - 1) 1, $\frac{1}{2}$ 2) $\frac{1}{2}, 1$ 3) $\frac{1}{2}, \frac{1}{2}$ 4) 1, 1
10. If X follows a binomial distribution with parameters $n = 8$ and $P = \frac{1}{2}$ then $P(|x - 4| \leq 2) =$
 - 1) $\frac{119}{128}$ 2) $\frac{9}{128}$ 3) $\frac{101}{128}$ 4) $\frac{11}{128}$
11. Consider 5 independent Bernoulli's trials each with probability of success p . If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval (AIE-2011)
 - 1) $\left[0, \frac{1}{2}\right]$ 2) $\left[\frac{11}{2}, 1\right]$
 - 3) $\left[\frac{1}{2}, \frac{3}{4}\right]$ 4) $\left[\frac{3}{4}, \frac{11}{12}\right]$
12. A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is (AIE-2007)
 - 1) $\frac{1}{729}$ 2) $\frac{8}{9}$ 3) $\frac{8}{729}$ 4) $\frac{8}{243}$



13. If on an average 1 vessel in every 10 is wrecked, the chance that out of 5 vessels expected 4 at least will arrive safely is

- 1) $\frac{4}{5}$ 2) $\frac{1}{2}$ 3) $\frac{2}{5}$ 4) $(\frac{14}{10})^{\frac{9}{5}}$

14. A man takes a step forward with probability 0.4 and backward with probability 0.6. The probability that at the end of eleven steps, he is just one step away from the starting point is

- 1) $462(0.24)^5$ 2) $368(0.24)^5$
3) $462(0.24)^6$ 4) $368(0.24)^6$

15. A die is thrown $2n+1$ times. The probability of getting 1 or 3 or 5 atmost n times is

- 1) $\frac{1}{2}$ 2) $\frac{1}{n}$ 3) $\frac{n}{(2n+1)}$ 4) $\frac{1}{(2n+1)}$

16. A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is (Mains - 2013)

- 1) $\frac{11}{3^5}$ 2) $\frac{10}{3^5}$ 3) $\frac{17}{3^5}$ 4) $\frac{13}{3^5}$

17. A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is

- 1) $8/729$ 2) $8/243$
3) $1/729$ 4) $8/9$

LEVEL - III - KEY

- 1) 3 2) 3 3) 3 4) 3 5) 2 6) 1
7) 1 8) 2 9) 1 10) 1 11) 1 12) 4
13) 4 14) 1 15) 1 16) 1 17) 2

LEVEL - III - HINTS

1. $f(p) = np(1-p) = np - np^2$;

$$p = \frac{1}{2}, q = \frac{1}{2}$$

2. $(\underline{2} \underline{2} \underline{2} \underline{3} \underline{3})$;

$$p = \frac{1}{2}, q = \frac{1}{2}, n = 5 ;$$

$$p(X=3) = {}^5C_3 \left(\frac{1}{2}\right)^5$$

3. $p(x \geq 4), n = 6, p = 2q$

4. $P = 0.05, n = 5, P(X \geq 1)$

5. $P = 0.05, n = 5, P(X = 0)$

6. $(3333), n = 4;$

$$P(X=4) = {}^4C_4 \left(\frac{1}{2}\right)^4$$

7. $\bar{x} + \sigma^2 = 24; \bar{x} \cdot \sigma^2 = 128, x^2 - \sigma^2 = 8$

8. $x : 3 \quad 2 \quad 1 \quad 0$
 $y : 0 \quad 1 \quad 2 \quad 3$

$$\text{Req.} = P(X=3) \cdot P(Y=0) + P(X=2) \cdot P(X=1) + P(X=1) \cdot P(X=2) + P(X=0) \cdot P(X=3)$$

9. $\mu = np, \sigma^2 = npq$

$$n = 2, p = \frac{3}{6}, q = \frac{3}{6}$$

10. $-2 \leq x - 4 \leq 2$

$$2 \leq x \leq 6$$

$$\text{Req.} = p(x=2) + p(x=3) + p(x=4) + p(x=5) + p(x=6)$$

11. $1 - p^5 \geq \frac{31}{32}$;

$$p \leq \frac{1}{2}$$

12. $p = \frac{1}{9}, n = 3$

13. $q = \frac{1}{4}, p = \frac{3}{4}, n = 5$

14. $p = 0.4, n = 11$, find $p(x=6) + p(x=5)$

15. $p(x=1) + p(x=3) + \dots + p(x=n)$

16. $n = 5, p = \frac{1}{3}, q = \frac{2}{3}$

17. Here $n = 3$,

$$p = 1/9,$$

$$q = 8/9, p(x=2)$$



LEVEL - IV

- 1. Statement-I : A fair coin is tossed 100 times. The probability of getting tails an odd number of times is $\frac{1}{2}$**

Statement-II : A fair coin is tossed 99 times. The probability of getting tails an odd number of times is $\frac{1}{2}$

Then which of the above statements are true.

- 1) only I 2) only II
3) both I and II 4) neither I nor II

- ## 2. For the binomial distribution

$(q+p)^n = \left(\frac{1}{2} + \frac{1}{2}\right)^{10}$, if $A=P(x=3)$, $B=P(x=6)$, $C=P(x=9)$ then the ascending order of A, B, C is

- 1) C, A, B 2) A, B, C 3) A, C, B 4) B, C, A

- 3. One hundred identical coins are thrown as each coin has the probability of getting head as p . Let x = number of coins showing heads then match the following conditions with p value.**

A) $P(x = 49) = P(x = 50)$ 1) $\frac{1}{2}$

B) $P(x = 48) = P(x = 52)$ 2) $\frac{51}{101}$

C) $P(\mathbf{x} = \mathbf{r}) = P(\mathbf{x} = \mathbf{n} - \mathbf{r})$ 3) $\frac{50}{101}$

4) $\frac{57}{100}$

The correct matching is

	<u>A</u>	<u>B</u>	<u>C</u>
1)	2	3	1
2)	4	2	3
3)	1	2	3
4)	3	1	1

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- 4. Observe the following statements**

Assertion (A) : X is binomial variate with parameters $2n + 1$ and $p = \frac{1}{2}$ then $P(x = \text{odd})$

$$\text{values}) = \frac{1}{2}$$

Reason (R) : If ${}^nC_r = C_r$ then

$C_1 + C_3 + C_5 + \dots = 2^{n-1}$. Then

- 1) A is false, R is true
- 2) A is true, R is true but $R \not\Rightarrow A$
- 3) A is true, R is true and $R \Rightarrow A$
- 4) A is true, R is false

LEVEL - IV - KEY

- 1) 3 2) 1 3) 4 4) 3

LEVEL - IV - HINTS

1. $P(x=1) + p(x=3) + \dots + p(x=99)$
2. $n=10, p=1/2, q=1/2$
3. $n=100, p(X=r) = n_{c_r} q^{n-r} p^r$
4. $A : \left(2n+1_{C_1} + 2n+1_{C_3} + \dots \right) \frac{1}{2^{2n+1}} = \frac{1}{2}$
